

Linear Regression Model for House Price Prediction

Name: Yashashvi Bisen
Course: Computer Science Engineering

Tool: Python (Google Colab)
Dataset: California Housing

1 Introduction

Machine Learning is a subset of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed. It is widely used in real-world applications such as recommendation systems, fraud detection, and price prediction.

In this project, a **Linear Regression model** is developed to predict house prices using the **California Housing dataset**. The project demonstrates the complete machine learning workflow, including data loading, exploration, preprocessing, model training, and evaluation.

2 Dataset Description

The **California Housing dataset** contains information collected from different districts in California. It is a standard benchmark dataset widely used in machine learning research.

Feature	Description	Type
Median Income	Median income of block group	Numeric
House Age	Median age of houses in block	Numeric
Average Rooms	Average number of rooms	Numeric
Average Bedrooms	Average number of bedrooms	Numeric
Population	Population of block group	Numeric
Average Occupancy	Average house occupancy rate	Numeric
Latitude	Geographic latitude coordinate	Numeric
Longitude	Geographic longitude coordinate	Numeric
Median House Value	Target: House price to predict	Target

3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify important patterns before building the model.

Methods Used:

- `.head()` — View the first few rows of sample data
- `.info()` — Understand data types and null values

- `.describe()` — Statistical summary (mean, std, min, max)
- Correlation heatmap — Analyze feature relationships with the target variable

■ **Key Observation:** A strong positive correlation was found between **Median Income** and house prices, making it the most influential predictor in the dataset. EDA helps in selecting relevant features and improving model performance significantly.

4 Methodology

The following pipeline was followed to build and evaluate the machine learning model:

1	Data Loading	Load California Housing dataset using Scikit-learn's built-in function
2	Data Exploration	Visualize distributions, correlations and statistical summaries
3	Preprocessing	Scale features using StandardScaler for better model performance
4	Train/Test Split	Divide data: 80% Training set and 20% Testing set
5	Model Training	Fit Linear Regression model on the training data
6	Prediction	Generate predictions on unseen test data
7	Evaluation	Measure performance using MAE, RMSE, and R ² Score

5 Model Used — Linear Regression

Linear Regression is a supervised machine learning algorithm used for predicting continuous values. It models the relationship between input features and the target variable by fitting a straight line (hyperplane in multiple dimensions) to the training data.

The model equation is:

$$\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n$$

Where $\hat{y}$ is the predicted value, β_0 is the intercept, and $\beta_1 \dots \beta_n$ are the feature coefficients learned during training.

6 Evaluation Metrics & Results

0.53 MAE	0.72 RMSE	0.65 R ² Score
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Metric	Value	Interpretation
Mean Absolute Error (MAE)	0.53	Average absolute difference between predicted and actual values
Root Mean Squared Error (RMSE)	0.72	Penalizes larger errors more; sensitive to outliers

R² Score	0.65	Model explains 65% of variance — moderate prediction accuracy
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Results and Visualization

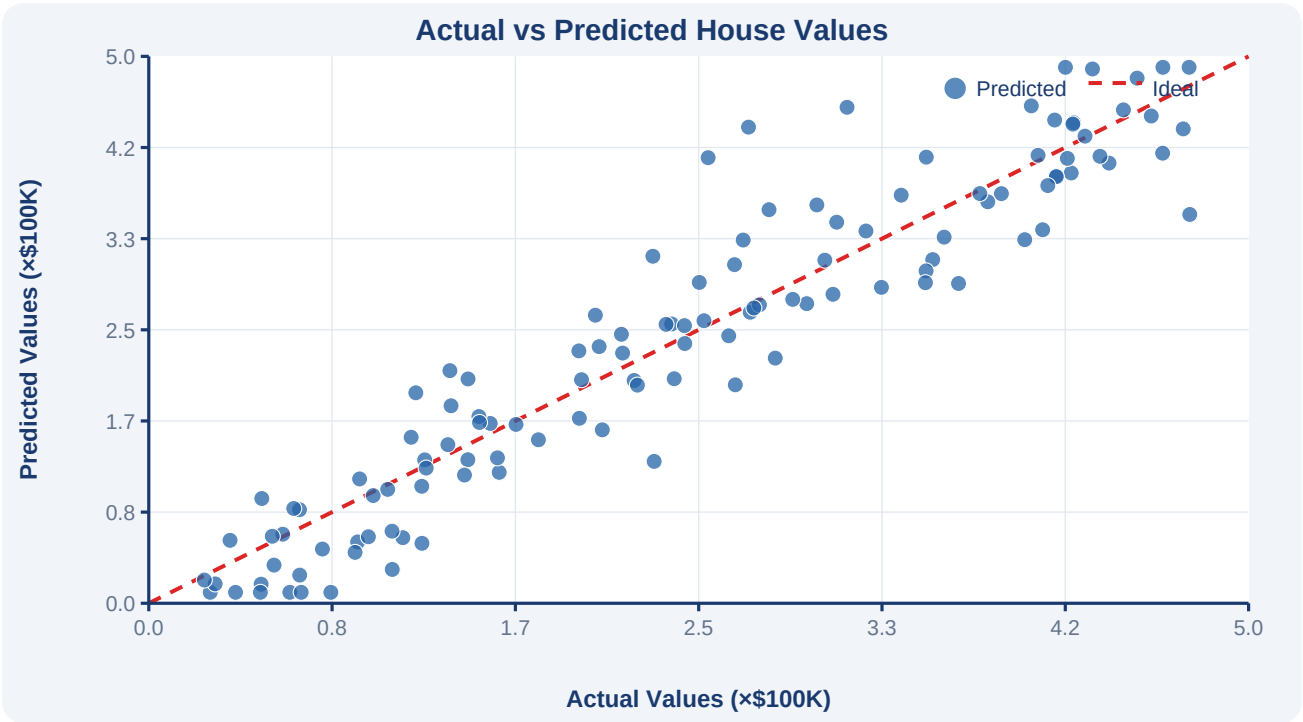


Fig 1. Scatter plot of Actual vs Predicted house values — points along the diagonal indicate accurate predictions.

■ **Observation:** Points closer to the diagonal line indicate better predictions. The spread pattern reveals some deviation for higher-value properties, suggesting scope for improvement with more advanced models.

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Improvement Ideas

The model performance can be enhanced by applying the following techniques:

■	Ensemble of decision trees; reduces overfitting and variance
■	Iteratively builds models to correct previous errors (XGBoost/LightGBM)
■	Create new meaningful features from existing ones (e.g. rooms per person)
■	GridSearchCV or RandomizedSearchCV to optimize model parameters
■	Identify and remove extreme values that distort the regression line
■	More training examples improve generalization and model accuracy

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Conclusion

This project successfully demonstrates the implementation of a **Linear Regression model** for predicting house prices using the California Housing dataset. The model achieves an **R² Score of 0.65**, providing a solid baseline performance.

The project covers the complete machine learning workflow — from data exploration and preprocessing to model training and evaluation — showcasing how ML can be applied to solve real-world prediction problems. With further improvements such as advanced algorithms and feature engineering, the accuracy can be enhanced significantly.

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References

- Scikit-learn Documentation — <https://scikit-learn.org/stable/>
- California Housing Dataset — Pace, R.K. & Barry, R. (1997)
- Machine Learning Tutorials — Géron, A. (2019). Hands-On ML with Scikit-Learn & TensorFlow
- Python ReportLab PDF Generation — <https://www.reportlab.com/>